

EXHIBIT 39

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U.S. Patent No. 10,707,167 Infringement Chart

Title: Contacts to semiconductor substrate and methods of forming same

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Abstract: An aspect of the invention includes a method for forming a contact in a dielectric layer over a semiconductor substrate. The method may comprise: forming a contact opening in a dielectric layer over the semiconductor substrate to expose an upper portion of the semiconductor substrate; depositing a first liner layer to conformally coat the contact opening; causing a portion of the first liner layer to diffuse into the upper portion of the semiconductor substrate to form a first intermix region at the upper portion of the semiconductor substrate; depositing a refractory metal layer over the first intermix region; and depositing a metal in the contact opening thereby forming the contact.

Claim	Tower '167 Accused Products
[1pre] A semiconductor structure including a contact to a semiconductor substrate, the contact comprising:	To the extent the preamble is limiting, devices manufactured by Tower's RF processes (including its 65nm RF process) (hereinafter, "Tower '167 Accused Products") comprise a semiconductor structure, as explained further in the limitations below.¹ For example, the Tower '167 Accused Products include the Qorvo QM81026 chip that, on information and belief, has substantially similar structure, function, operation, and implementation with respect to the claims of the '167 patent as any other chip manufactured by Tower using its 65nm processes.

¹ GlobalFoundries reserves the right to identify to identify different or additional Tower '167 Accused Products and amend or supplement these charts after further investigation and discovery, claim construction, or other meriting circumstances. Such supplemental information may include, but is not limited to, source code, data sheets, design specifications, deposition testimony, testing information, reference designs implementation and utilization information, and/or schematics.

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Claim	Tower '167 Accused Products
	
[1a] a contact opening in a dielectric layer over the semiconductor substrate;	The Tower '167 Accused Products comprise a contact opening in a dielectric layer over the semiconductor substrate. For example, the Tower '167 Accused Products includes a contact opening in a dielectric layer over the semiconductor substrate.

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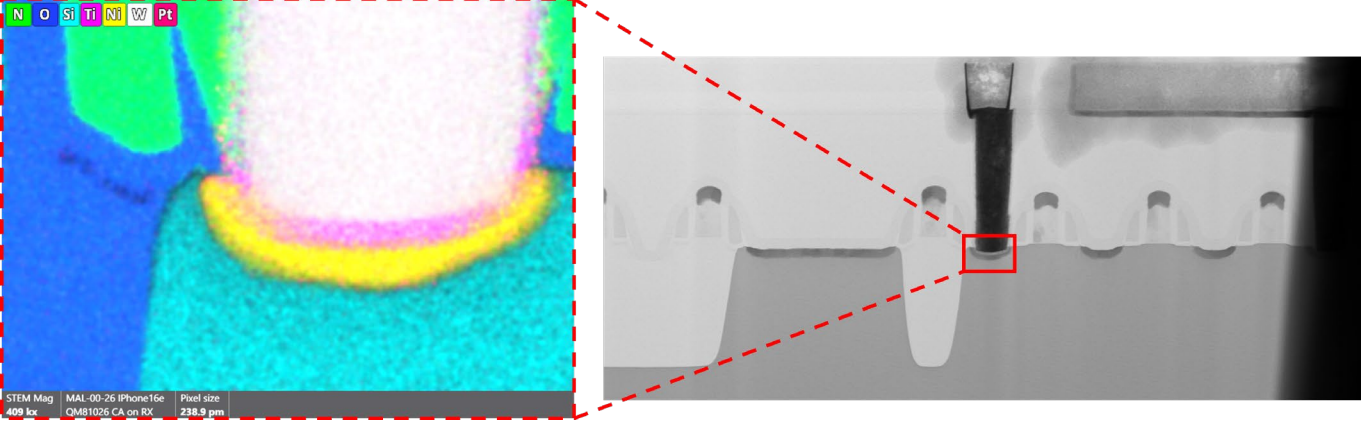
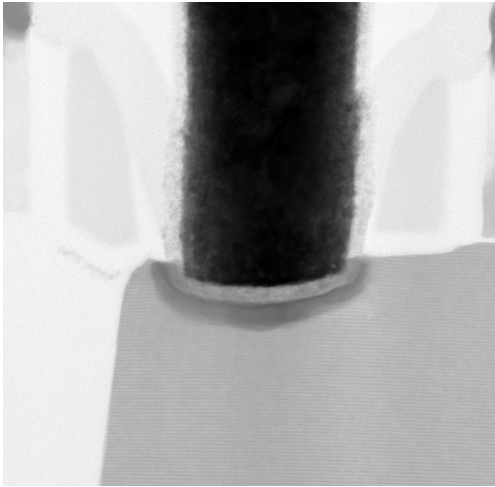
Claim	Tower '167 Accused Products
	 The figure consists of two parts. On the left is an EDS map showing elemental distribution with a color scale for N, O, Si, Ti, Ni, W, and Pt. A red dashed box highlights a specific region. On the right is an SEM image of a contact opening with a red box indicating the area shown in the EDS map. Technical data at the bottom left of the EDS map includes: STEM Mag 409 kx, MAL-00-26 iPhone16e, QM81026 CA on RX, and Pixel size 238.9 pm.
[1b] a first liner layer substantially coating a sidewall of the contact opening;	The Tower '167 Accused Products comprise a first liner layer substantially coating a sidewall of the contact opening. For example, the '167 Accused Products include a first liner layer substantially coating a sidewall of the contact opening.  This SEM image provides a magnified view of a contact opening. A dark, textured layer is visible coating the sidewall of the opening, which corresponds to the 'first liner layer' mentioned in the claim and description.

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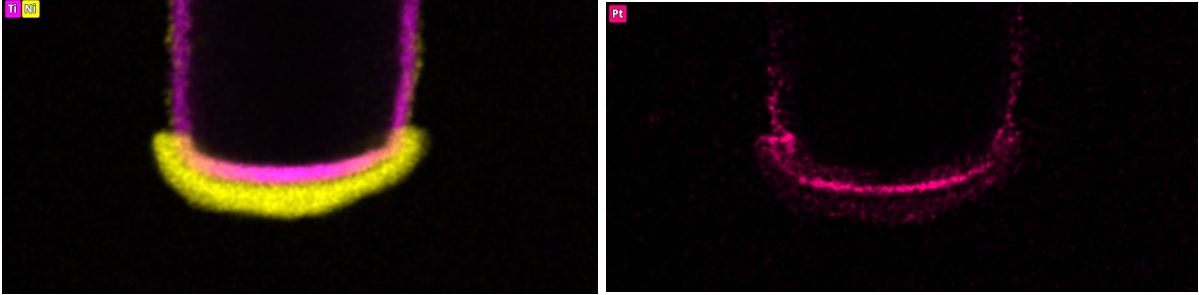
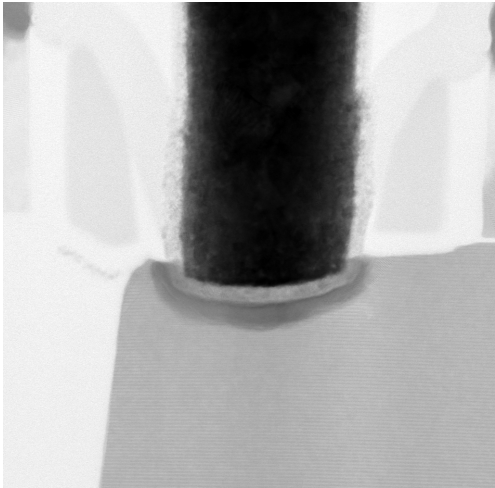
Claim	Tower '167 Accused Products
	
[1c] a refractory metal layer substantially coating the first liner layer in the contact opening;	The Tower '167 Accused Products comprise a refractory metal layer substantially coating the first liner layer in the contact opening. For example, the '167 Accused Products include a refractory metal layer substantially coating the first liner layer in the contact opening.  For example, the materials analysis below shows a refractory metal layer substantially coating the first liner layer in the contact opening.

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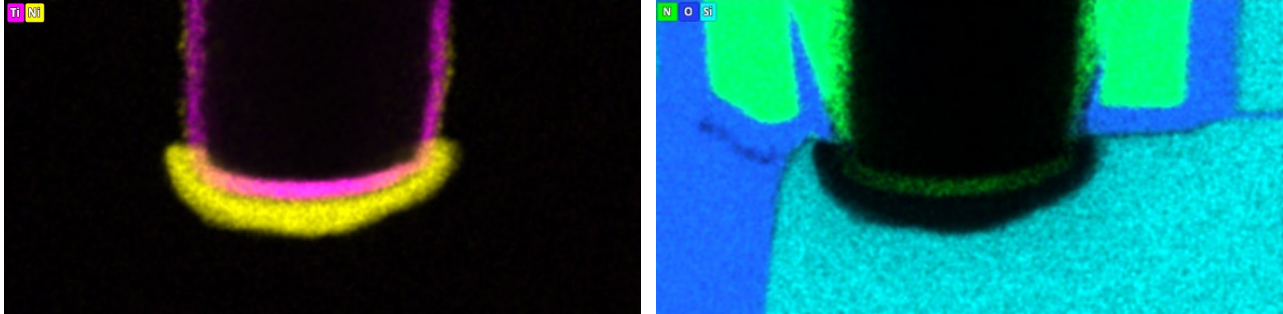
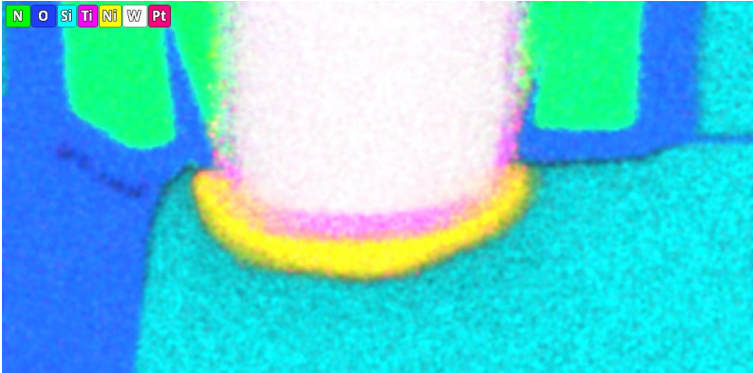
Claim	Tower '167 Accused Products
	
[1d] a metal substantially filling the contact opening; and	The Tower '167 Accused Products comprise a metal substantially filling the contact opening. For example, the materials analysis below illustrates a metal substantially filling the contact opening. 
[1e] an intermix region at an upper surface of the semiconductor substrate in the contact opening, the intermix region having a composition including a material used in the first liner layer, a material used in the refractory metal layer, and a material used in the semiconductor substrate	The Tower '167 Accused Products comprise an intermix region at an upper surface of the semiconductor substrate in the contact opening, the intermix region having a composition including a material used in the first liner layer, a material used in the refractory metal layer, and a material used in the semiconductor substrate. For example, the materials analysis below shows the presence of an intermix region at an upper surface of the substrate. The green arrow in the image of the contact stack below corresponds to the green arrow at the top of the ensuing materials analysis chart.

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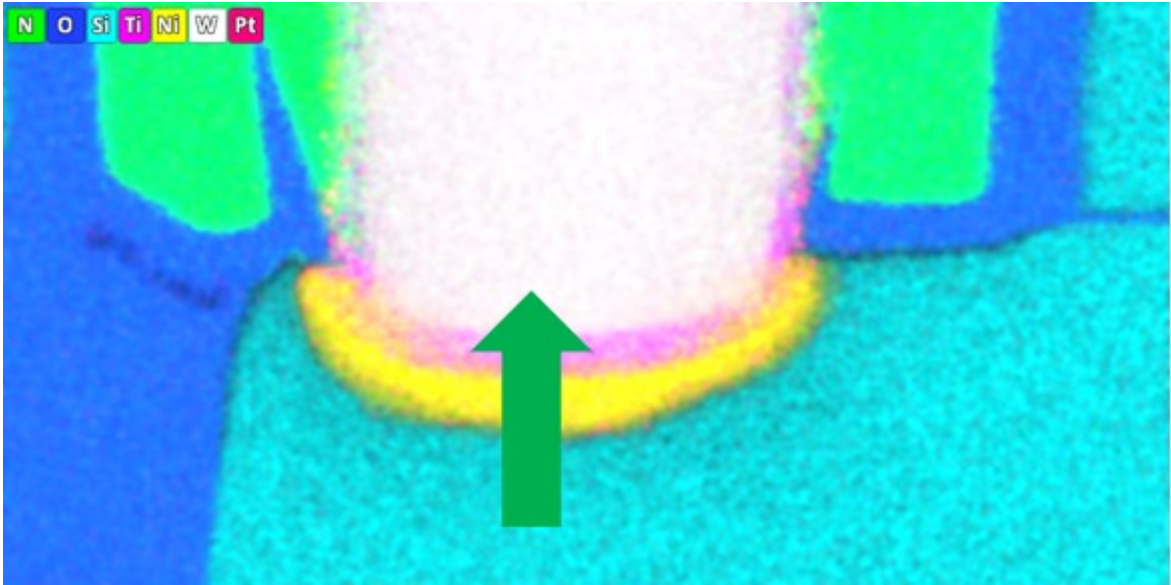
Claim	Tower '167 Accused Products
liner layer, a material used in the refractory metal layer, and a material used in the semiconductor substrate;	An EDS-EDS image showing a cross-section of a semiconductor device. The image displays various layers and regions in different colors. A legend at the top left identifies the elements: N (blue), O (red), Si (green), Ti (yellow), Ni (cyan), W (magenta), and Pt (black). A green arrow points to a specific region in the center of the image, which is a mix of colors, indicating an intermix region. The region is surrounded by a blue layer (likely Si) and a yellow layer (likely Ti). The background is a mix of blue and red, indicating a substrate. As shown, the intermix region covered by the green arrow in the image above has a composition that includes a material used in the first liner layer, the refractory metal layer, and the semiconductor substrate.

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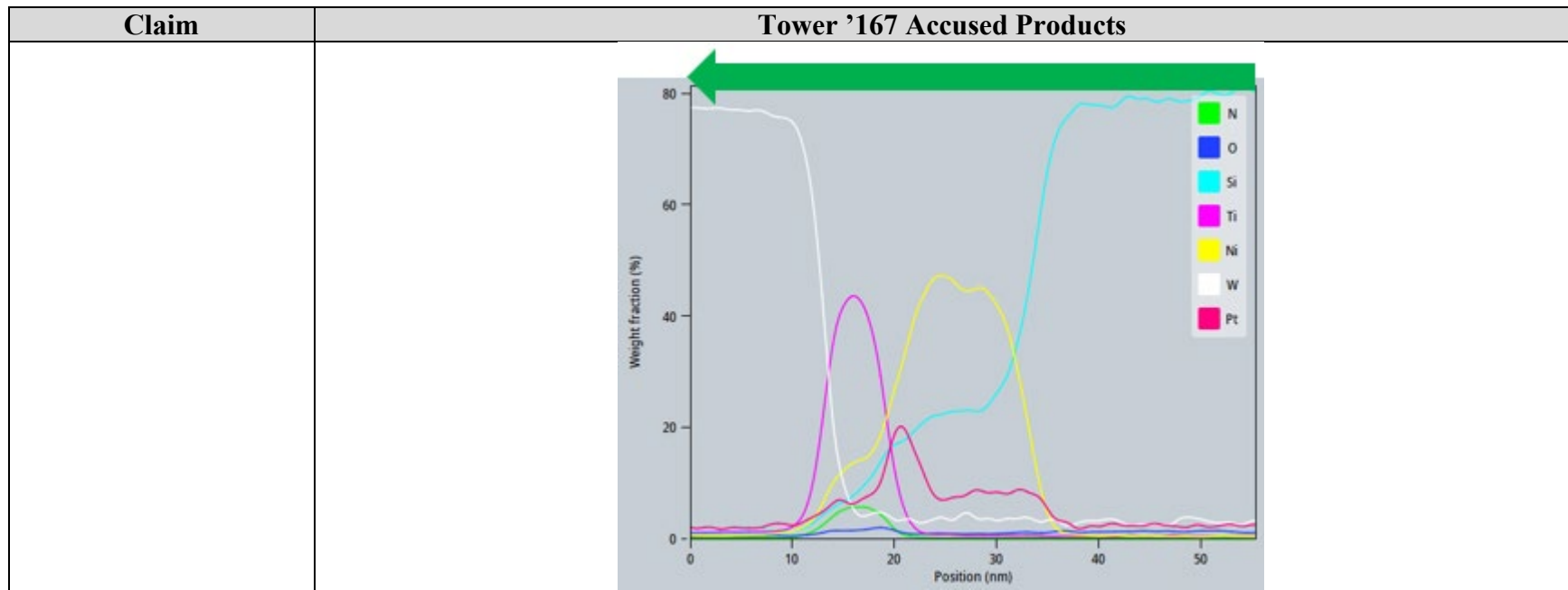


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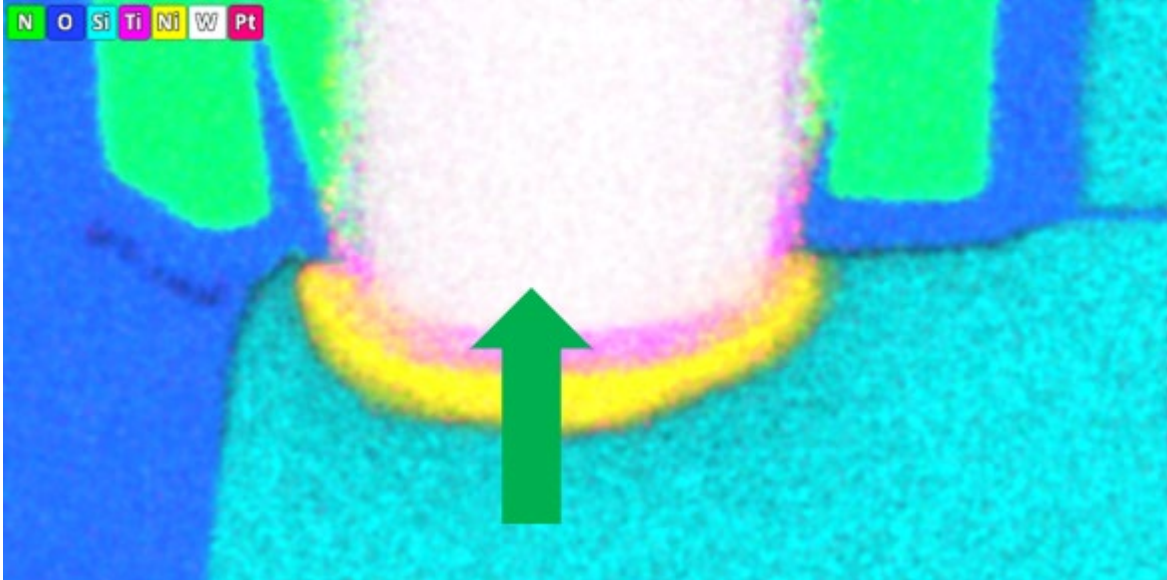
Claim	Tower '167 Accused Products
[1f] wherein the refractory metal layer includes a multi-layer having at least a first refractory metal material and a second refractory metal material, the first refractory metal material substantially separating the first liner layer and the second refractory metal material, the second refractory metal material substantially separating the first refractory metal material and the metal.	In the Tower '167 Accused Products, the refractory metal layer includes a multi-layer having at least a first refractory metal material and a second refractory metal material, the first refractory metal material substantially separating the first liner layer and the second refractory metal material, the second refractory metal material substantially separating the first refractory metal material and the metal. For example, the refractory metal layer includes a first refractory metal material and a second refractory metal material, as can be seen in the image below and the ensuing material analysis chart. As shown in the material analysis chart, the first refractory metal material substantially separates the first liner layer and the second refractory metal material. Likewise, the second refractory metal material substantially separates the first refractory metal material and the metal. 

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